

The Black-Scholes Model

(5. The Greeks)

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

The BSM Greeks

Greeks are sensitivities of Options w.r.t to various Input parameters. Greeks are essential for risk calculations, hedging and local valuations or PnL (Taylor Series!)

Numerator / Denominator	Spot (S)	Volatility (σ)	Time (t)	Interest rate (r)
Value (V)	Delta (Δ)	Vega (ν)	Theta (θ)	Rho (ρ)
Delta (Δ)	Gamma (Γ)	Vanna	Charm	
Vega (ν)	Vanna	Vomma/Volga	Veta	
Theta (θ)	Charm	Veta		
Gamma (Γ)	Speed	Zomma	Color	
Vomma		Ultima		

$$\frac{\partial V}{\partial S} \rightarrow$$

$$\frac{\partial \Delta}{\partial S}$$

$$= \frac{\partial^2 V}{\partial S^2}$$

$$\frac{\partial \nu}{\partial S} = \frac{\partial^2 V}{\partial S \partial \sigma}$$

$$\frac{\partial \nu}{\partial \sigma} = \frac{\partial^2 V}{\partial \sigma^2}$$

► Why some of the values appear symmetrically in the grid ?

The Black Scholes in form of Greeks

The Greeks appear naturally in the Black Scholes equation

$$\rightarrow \frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

$$\Theta + \frac{1}{2} \sigma^2 S^2 \Gamma + rS \Delta - rV = 0$$

Generalized Black Scholes

Black-Scholes Option pricing formula can be generalized to cater to various types of equity options by introducing a **cost of carry**(b) term.

$$V = \omega S e^{(b-r)T} N(\omega d_1) - \omega K e^{-rT} N(\omega d_2)$$

$$d_1 = \frac{\ln\left(\frac{S}{K}\right) + \left(b + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

where $\omega = +1$ for Call Option and $\omega = -1$ for Put options

$$F = S e^{(b-r)T}$$

- ▶ $b = r$ gives stock option model (Black Scholes Merton 1973)
- ▶ $b = r - D$ gives stock option model with continuous dividend yield D (Merton 1973)
- ▶ $b = r + (c) - (y)$ gives the commodity option with storage cost (C) and convenience yield (y)
- ▶ $b = 0$ gives option on futures model (Black 1976)
- ▶ $b = r_d - (r_f)$ gives option on currency (Garman and Kohlhagen 1983)

Delta (Δ)

Delta is the option sensitivity to small changes in the spot price (S)

Call

$$\Delta_{\text{call}} = \frac{\partial c}{\partial S} = e^{(b-r)T} N(d_1)$$

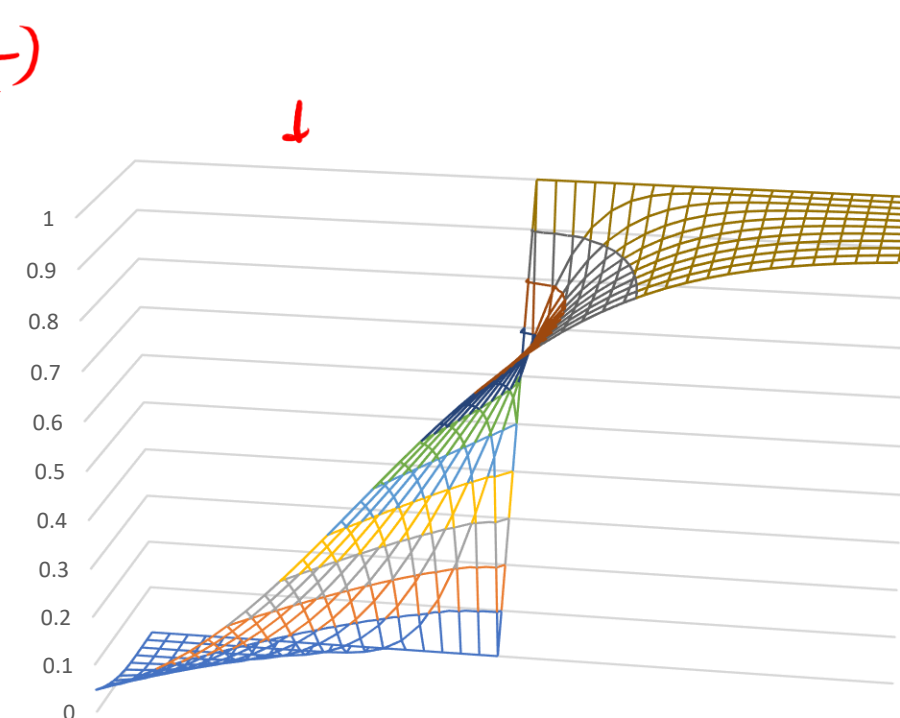
$b=r$

Put

$$\Delta_{\text{put}} = \frac{\partial p}{\partial S} = e^{(b-r)T} [N(d_1) - 1]$$

For call options, the delta is +ve. It's closer to 0 if the option is deep-OTM and closer to 1 if it's deep ITM. For put options, the delta is -ve.

The value of delta evolves smoothly when the time to maturity is large but shows spikes closer to maturity.



Delta (Exercises)

1. Derive the result $S_0 \varphi(d_1) = K e^{-rT} \varphi(d_2)$. ✓
2. Derive the result $\Delta_{\text{call}} = \frac{\partial c}{\partial S} = N(d_1)$ ✓
3. Derive a relation between Δ_{call} and Δ_{put}
4. Can the delta of a call option be more than 1? ✓

$$\frac{e^{(b-r)} \cdot N(d_1)}{>1}$$

$$r < 0$$

$$c = S_0 N(d_1) - K e^{-rT} N(d_2)$$

$$\Delta = \frac{\partial c}{\partial S_0} = N(d_1)$$

$$\begin{aligned} \textcircled{N} &\rightarrow \text{CDF} \\ \rightarrow \Phi &\rightarrow \text{CDF} \\ \rightarrow \phi &\rightarrow \text{PDF} \end{aligned}$$

Vega (v)

Vega is the first order sensitivity of option price with respect to the asset volatility.

Call and Put

$$v = \frac{\partial c}{\partial \sigma} = S_0 e^{(b-r)T} \sqrt{T} \phi(d_1)$$

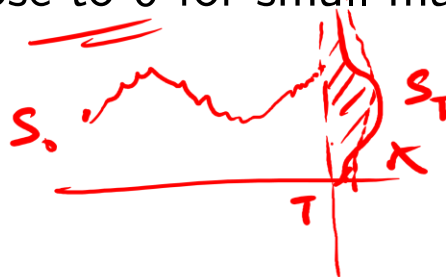
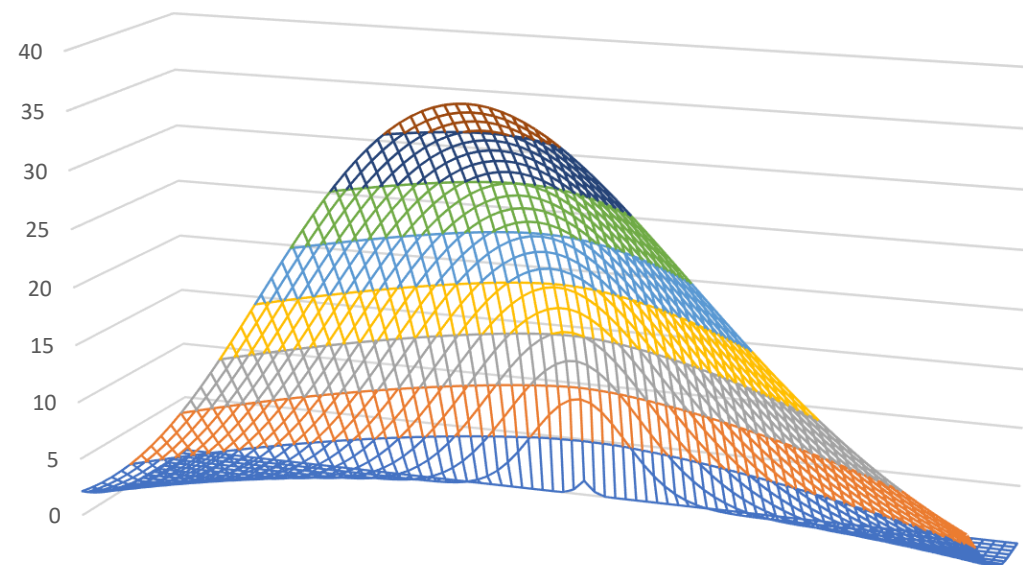
$$P_0 + S_0 = C_0 + ke^{-rT}$$

$$v_{\text{put}} = v_{\text{call}}$$

Vega for call and put are always positive and the same.

It's higher for long dated options and ATM forward.
(max value is obtained at $d_1 = 0$)

It's close to 0 for small maturities or deep ITM/OTM



$$S_T - K$$

$$K - S_T$$

$$S_0 = K$$

$$K = S_0 e^{rT}$$

Theta (Θ)

Theta is the first order sensitivity of option price with respect to time.

Call

$$\Theta = \frac{\partial c}{\partial t} = -\frac{\partial c}{\partial T} = -\frac{S_0 e^{(b-r)T} \varphi(d_1) \sigma}{2\sqrt{T}} - (b-r)S_0 e^{(b-r)T} N(d_1) - rKe^{-rT} N(d_2)$$

Put

$$\Theta = \frac{\partial c}{\partial t} = -\frac{\partial c}{\partial T} = -\frac{S_0 e^{(b-r)T} \varphi(d_1) \sigma}{2\sqrt{T}} + (b-r)S_0 e^{(b-r)T} N(-d_1) + rKe^{-rT} N(-d_2)$$

Theta is generally negative (time decay), but with **deep-in-the-money puts**, theta can be +ve.

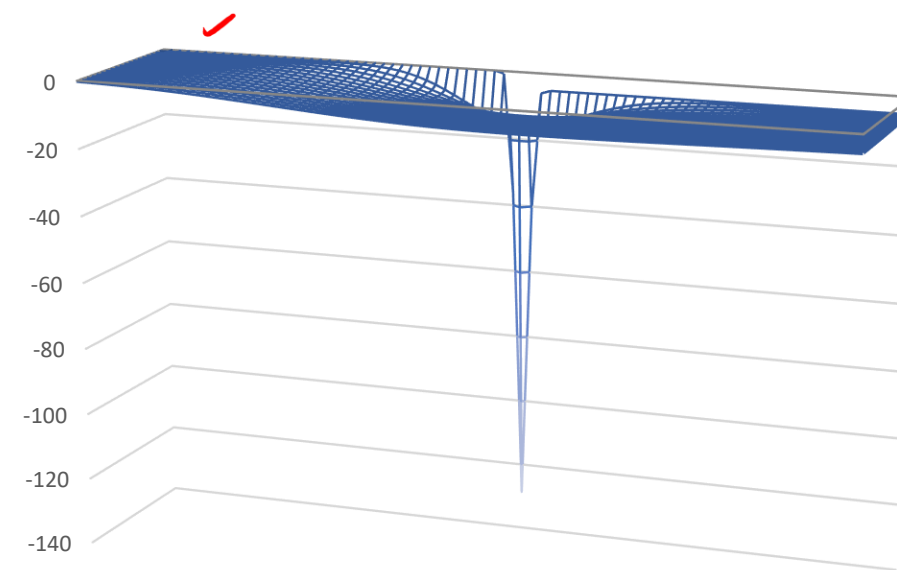
It's small for long maturities or deep ITM/OTM

Theta spikes near maturity for ATM options

Show that $\Theta_{\text{call}} < \Theta_{\text{put}}$

$t \rightarrow +\uparrow$ $\frac{T-P}{T \downarrow}$

$\frac{K-S}{T \downarrow}$



Rho (ρ)

Rho is the first order sensitivity of option price with respect to risk-free rate.

Call $\rho_{\text{call}} = \frac{\partial c}{\partial r} = \underline{\underline{TKe^{-rT}N(d_2)}}$

Put $\rho_{\text{put}} = \frac{\partial p}{\partial r} = \underline{\underline{-TKe^{-rT}N(-d_2)}}$



Rho is +ve for call options and -ve for put options.



$\boxed{e^{-rT}}$

Call $(S - K)$

$K e^{-rT}$

Put $(K - S)$

Gamma (Γ)

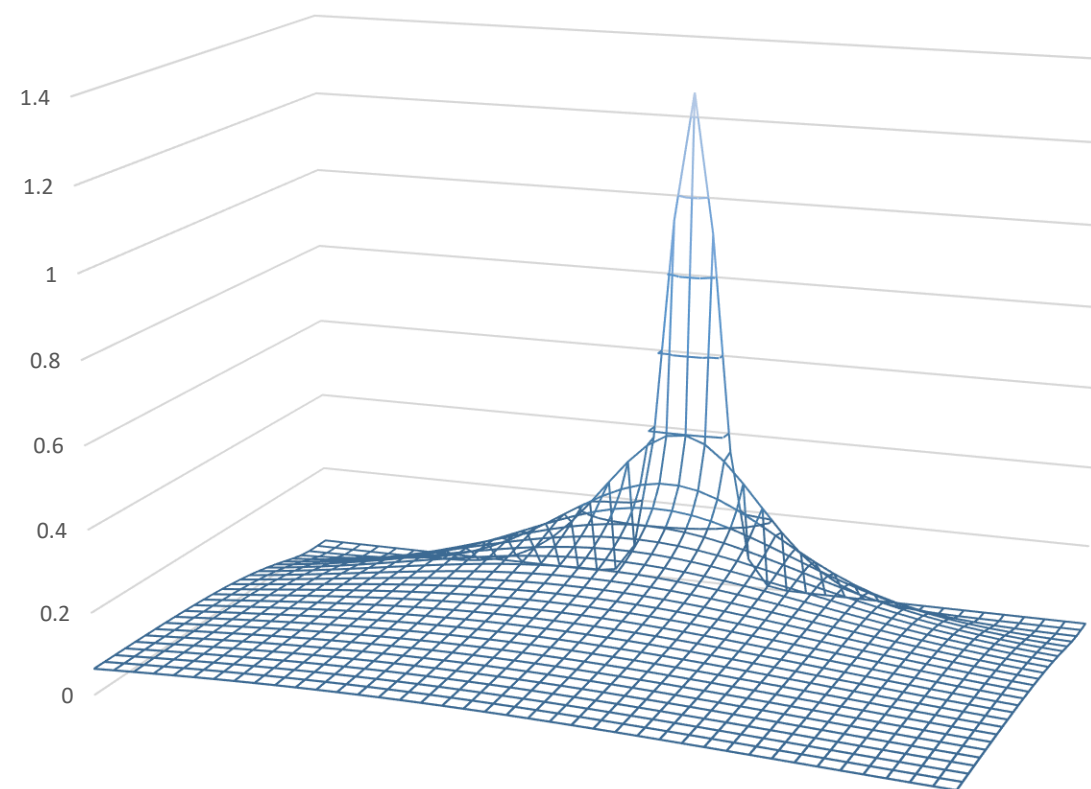
Gamma is the second order sensitivity of option price with respect to spot price.

Call and Put $\Gamma = \frac{\partial^2 c}{\partial S^2} = \frac{\partial^2 p}{\partial S^2} = \frac{e^{(b-r)T} \boxed{\varphi(d_1)}}{S\sigma\sqrt{T-t}}$

Handwritten notes: $t \rightarrow T$, $d_1 = 0$, $\varphi(0)$

Gamma is +ve and the same for call and put.

Gamma is highest for ATM and near expiry.



Application (PnL)

Today the stock price is trading at \$100. Assume risk free rate = 5%, time to maturity = 1 year, volatility = 20%, Strike = \$100. Calculate the Black Scholes Call Option price. Tomorrow if the stock price appreciates by 5% and the volatility appreciates by 2%, calculate the PnL using

- 1) Full Revaluation ✓
- 2) Local Valuation ✓

Hint : Taylor Series

$$\cancel{f(x + \epsilon)} \approx \cancel{f(x)} + \cancel{f'(x)} \epsilon + \frac{1}{2} \cancel{f''(x)} \epsilon^2$$

Application (Hedging I)

Suppose you have a call option referencing 100 shares of stock. You want to create a delta neutral portfolio. Calculate how many units of hedging instrument you need to create a delta-neutral portfolio where the hedging instrument is

1) Stocks ✓

c, S, F

2) Futures ✓

$$1) \pi_1 = \check{n}_1 \check{C} + \check{n}_2 \check{S}, \quad \frac{\partial \pi_1}{\partial S} = 0$$

$$= n_1 \Delta_{call} + n_2 = 0$$

$$\textcircled{n_2} = \frac{-(n_1) \cdot \Delta_{call}}{1}$$

2)

$$2) \pi_2 = n_1 C + n_2 F$$

$$\frac{\partial \pi_2}{\partial S} = 0, \quad \frac{\partial \pi_2}{\partial F} = 0$$

$$n_1 \frac{\partial C}{\partial F} + n_2 = 0$$

$$\Rightarrow n_2 = -n_1 \cdot \frac{\partial C}{\partial F} = -n_1 \times \frac{\partial C}{\partial S} \cdot \frac{\partial S}{\partial F}$$

$$= -n_1 \times \Delta_{call} \times e^{-rT}$$

$$\Delta_F = \Delta_S \times e^{-rT}$$

$$F = S e^{rT}, \quad \frac{\partial F}{\partial S} = e^{rT}$$

Application (Hedging II)

An Investor sold 100 units of 33-strike call option. From the information below find the units of stock and 35-strike call option investor should buy/sell to maintain a delta and gamma hedged portfolio.

	$k_1 = 33$ -strike Call	$k_2 = 35$ -strike Call
delta	✓ 0.6986	✓ 0.5217
gamma	✓ 0.07584	✓ 0.06245

$$V = -n_1 \cdot C(k_1) + n_2 C(k_2) + n_3 \cdot S$$

$$\Delta = 0 \Rightarrow -n_1 \Delta_{k_1} + (n_2) \Delta_{k_2} + (n_3) = 0$$

$$\Gamma = 0 \Rightarrow -n_1 \Gamma_{k_1} + (n_2) \Gamma_{k_2} = 0$$

$n_2, n_3 \ll$

Dual Deltas (Probability deltas)

Dual deltas are defined as first order sensitivity of option to strike price. It may sound illogical to talk about changes in strike price as strike price remains constant for the contract. However, this can be interpreted as the difference in price of another option whose strike is slightly different than the first.

$$\underbrace{V_1}_{\text{at } K} \rightarrow \underbrace{(K + \epsilon)}_{\text{at } V_2}$$

Show that the dual deltas are given as

$$\frac{S - K}{K - S}$$

$$\left(\frac{\partial c}{\partial K} \right) = -e^{-rT} N(d_2)$$

$$e^{-rT} N(d_2)$$

$$\frac{\partial p}{\partial K} = e^{-rT} N(-d_2)$$

$$\begin{cases} 1, & S_T > K \\ 0, & S_T \leq K \end{cases}$$


$$\begin{aligned} V &= \mathbb{E}^Q \left(e^{-rT} \times 1_{S_T > K} \right) \\ &= e^{-rT} \times Q(S_T > K) \\ &= e^{-rT} \times N(d_2) \end{aligned}$$

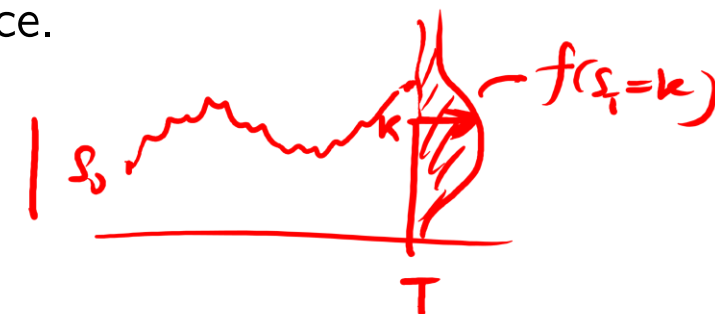
Since $N(d_2)$ is the risk-neutral probability of exercising the call option, the strike deltas or dual deltas simply give the discounted risk-neutral probability.

Show that the call strike delta represents the value of **digital call** (in absolute terms) which is also related to a **tight bull spread**.

Strike Gamma (Risk Neutral Density)

The risk-neutral density of the stock can be retrieved by differentiating the option price twice. In other words, the strike gamma gives the risk-neutral density of the underlying price.

$$\frac{\partial^2 c}{\partial K^2} = \frac{\partial^2 p}{\partial K^2} = \frac{e^{-rT} \phi(d_2)}{K \sigma \sqrt{T}} \geq 0$$



Show that the above expression is in fact the discounted risk-neutral density of stock at $S_T = K$ which is also related to a tight butterfly spread around K (Breeden-Litzenberger)

Density of lognormal distribution is $f(x) = \frac{1}{x \sigma \sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right)$